

Rise of NPUs in the Age of AI

A little over a decade ago, if you had mentioned Neural Processing Units (NPUs) at a tech conference, you'd likely have been met with blank stares or polite nods from an audience trying to decipher your tech jargon. Fast forward to today, and NPUs have become a staple in the silicon architecture of almost every major chip maker, from Intel to NVIDIA, Qualcomm to MediaTek, and even Samsung. So, what changed? Why are NPUs now a critical component in our devices, and what does this mean for the future?

Let's rewind to the early 2010s. The tech landscape was dominated by CPUs and GPUs, with the former handling general-purpose computing tasks and the latter excelling in parallel processing, crucial for gaming and graphics-intensive applications. But as artificial intelligence (AI) started to permeate every facet of technology, the limitations of CPU-GPU duopoly became apparent. Enter NPUs, specialised processors designed to accelerate machine learning tasks and handle the massive parallelism required by neural networks.

Think of an NPU as a highly specialised chef in a restaurant kitchen. While the CPU is the head chef, and the GPU is the sous-chef, the NPU is the pastry chef, meticulously focused on creating the perfect dessert with a creative touch. Each plays a crucial role, but the NPU's specialised skills make it indispensable for specific tasks like AI.

In fact, the trajectory of NPUs mirrors the rise of AI. As AI moved from research labs to everyday applications, the demand for efficient processing of AI tasks skyrocketed. The initial wave of AI applications, like image and voice recognition, required immense computational power. Traditional processors were not optimised for these tasks, leading to several inefficiencies.

For example, consider the rise of voice assistants like Siri, Alexa, and Google Assistant. These services rely heavily on natural language processing (NLP), a subset of AI that interprets and responds to human speech. Early versions of these assistants were sluggish. With the integration of NPUs, these tasks became real-time interactions that felt more natural and intuitive.

Smartphones are another perfect example. Released in 2017, the iPhone X featured Apple's A11 Bionic chip, which included a dedicated "Neural Engine." This NPU was specifically



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Jayesh Shinde
Executive Editor



Let me know your thoughts on this column at:
jayesh.shinde@digit.in |
@jshinde

designed to accelerate AI tasks, enabling features like Face ID and Animoji to execute instantaneously. Similarly, Google's Pixel phones leverage the Pixel Visual Core, an NPU that enhances image processing capabilities, resulting in stunning photos with advanced computational photography techniques.

In the broader tech ecosystem, NPUs have revolutionised various industries. Autonomous vehicles rely on NPUs to process data from sensors and cameras in real-time, making split-second decisions critical for safety. Healthcare applications use NPUs to analyse medical images, assisting doctors in diagnosing conditions with higher accuracy. Even in entertainment, NPUs power recommendation engines on streaming platforms, tailoring content suggestions to individual preferences. From relative obscurity, NPUs are driving the on-device AI boom – a fact that hasn't been lost on chip and device manufacturers.

The integration of NPUs into System on Chips (SoCs) by major players like Intel, Qualcomm, NVIDIA, MediaTek, and Samsung signifies a broader shift towards AI-centric computing. These companies recognize that the future of technology hinges on AI, and NPUs are the key to unlocking its full potential. By embedding NPUs into their SoCs, they ensure that devices are equipped to handle the growing demands of AI applications, from edge computing to cloud services. As a result, NPUs are at the heart of every conversation from AI-powered smartphones to the future of AI PCs.

This leaves no doubt in my mind that the evolution of NPUs is only going to accelerate. As AI models become more complex, the need for even more specialised and powerful NPUs will grow. We can expect future NPUs to offer greater efficiency, lower power consumption, and enhanced capabilities. Innovations in chip design, such as integrating NPUs with other specialised processors like Tensor Processing Units (TPUs), will further propel AI advancements.

At the same time, the democratisation of AI tools and frameworks means that NPUs will become accessible to a broader range of developers and applications. Imagine a world where every device around you, is equipped with an NPU, seamlessly processing AI tasks to offer smarter, more intuitive experiences. **d**